

Tipu Sultan

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Google Scholar — LinkedIn — Website — GitHub — ResearchGate — ORCID

EDUCATION

Saint Louis University

St. Louis, MO

Ph.D. in Aerospace and Mechanical Engineering, GPA: 4.00/4.00

Expected April 2027

- Award: Fully Funded Ph.D. Program.

Fordham University

New York, NY

M.Sc. in Data Science, GPA: 3.72/4.00

May 2024

- Award: GSAS Centennial Scholarship

Kyungsung University

Busan, South Korea

B.Sc. in Mechatronics Engineering, GPA: 4.17/4.50

Feb 2022

- Award: Merit-Based Scholarship (Ranked 1st).

SKILLS

Python, SQL, PyTorch, Hugging Face Transformers, scikit-learn, NumPy, pandas, SciPy, signal processing for EEG/EMG/eye-tracking (filtering, resampling, artifact handling, synchronization), multimodal transformers & fusion, self-supervised learning, ROS2/MoveIt (Kinova Gen3), Linux/HPC (Slurm), Docker.

EXPERIENCE

Ph.D. Research Assistant — Saint Louis University, MechaRithm Lab

St. Louis, MO

Robot learning, RGB-D perception, and reliable manipulation

Jun 2024 — Present

- Develop a video-to-robot imitation-learning pipeline that converts human RGB-D demonstrations into robot-compatible trajectories using hand-pose estimation, depth processing, 3D transformations, trajectory retargeting, and inverse kinematics.
- Build an RGB-D perception pipeline combining SAM2-based hand and object segmentation, camera calibration, temporal depth fusion, and 3D back-projection, increasing valid object observations from 52.1% to 65.7% across 7,167 frames.
- Construct a long-tail benchmark with 570 corruption conditions and 27,360 frames covering depth dropout, occlusion, blur, motion blur, and RGB-depth misalignment; develop a reliability estimator achieving 0.782 AUROC.
- Develop and evaluate manipulation methods for five assistive tasks on a Kinova Gen3 robot, combining imitation learning, uncertainty estimation, and feasibility checks for safe action execution.
- Build reproducible PyTorch pipelines for multimodal training, ablation studies, participant-independent evaluation, and large-scale auditing of 1,099 RGB-D demonstrations and 154,844 frames.
- *Tech:* Python, PyTorch, OpenCV, SAM2, RGB-D perception, imitation learning, diffusion policies, ROS 2, MoveIt, MuJoCo, Kinova Gen3, and Robotiq gripper.

Data Analyst Intern — Authentic Brands Group

New York, NY

Analytics and Visualization (SQL, Tableau, Python)

Jan 2024 — Apr 2024

- Performed EDA with SQL and Python (pandas) across Salesforce, database, and Excel sources to surface trends.
- Built interactive Tableau dashboards; integrated Salesforce + Excel via Tableau Prep to streamline reporting.
- Automated ETL and data quality checks in Python (pandas, NumPy); optimized SQL; delivered weekly KPI reports (Excel/VBA).

Research Assistant — Fordham University

New York, NY

Multimodal AI for clinical NLP and medical imaging

2022 — 2024

- Led publication-focused research on multimodal clinical AI, resulting in IEEE Access papers including MediGPT (medical LLM benchmarking), MemesViTa (multimodal fusion), and SumGPT (radiology report summarization), plus GastroVRG (GI early screening).
- Built end-to-end radiology summarization system (image + report): dataset construction (pairing, cleaning, PHI-safe processing), multimodal alignment, and clinical-style summarization using transformer-based encoders and encoder-decoder/LLM models.
- Developed and compared multimodal fusion strategies (cross-attention, late fusion) and ran rigorous ablations, error analysis, and reliability evaluation (calibration/ECE) to ensure trustworthy clinical predictions.
- Created reproducible training + evaluation pipelines (config-driven runs, mixed precision, experiment tracking) and delivered strong baselines with clear empirical conclusions for papers and revisions.
- *Tech:* Python, PyTorch, Hugging Face, MONAI, scikit-learn; SQL; Docker; OpenCV; clinical text processing (spaCy); DICOM/NIfTI workflows.

Research Assistant – Kyungshung University

Busan, South Korea

Research on Switched Reluctance Machine Design and Optimization

May 2021 – Jul 2022

- Conducted research on the preliminary design and optimization of a Switched Reluctance Machine (SRM) for Flywheel Energy Storage System (FESS) applications, focusing on performance and efficiency.
- Developed and analyzed control strategies for the Switched Reluctance Motor/Generator (SRM/G) used in FESS, ensuring reliable and efficient operation under varying load conditions.
- Contributed to the design and control system modeling, addressing key considerations for enhancing system stability, energy efficiency, and durability.
- Applied simulation tools and analytical techniques to evaluate the feasibility and performance of the SRM/G in energy storage applications.

PUBLICATIONS

Journal Articles (Published)

- [J6] **T. Sultan**, G. Liu, P. Sikorski, S. Alshathri, W. El-Shafai, and M. Babaiasl. *NeuroFusion-Trans: A Novel Transformer-Based EEG-EMG Fusion Model for Assistive Robotics*. *IEEE Access*, 2025. DOI: 10.1109/ACCESS.2025.3577996.
- [J5] **T. Sultan**, G. Liu, P. Sikorski, and M. Babaiasl. *TransNN-MHA: A Transformer-Based Model to Distinguish Real and Imaginary Motor Intent for Assistive Robotics*. *IEEE Access*, 2025. DOI: 10.1109/ACCESS.2025.3567908.
- [J4] **T. Sultan**, M.A.T. Rony, M.S. Islam, S. Alshathri, and W. El-Shafai. *SumGPT: A Multimodal Framework for Radiology Report Summarization to Improve Clinical Performance*. *IEEE Access*, 2025. DOI: 10.1109/ACCESS.2025.3528335.
- [J3] M.S. Islam, M.A.T. Rony, and **T. Sultan**. *GastroVRG: Enhancing Early Screening in Gastrointestinal Health via Advanced Transfer Features*. *Intelligent Systems with Applications*, 23, 200399, 2024. DOI: 10.1016/j.iswa.2024.200399.
- [J2] M.A.T. Rony, M.S. Islam, **T. Sultan**, S. Alshathri, and W. El-Shafai. *MediGPT: Exploring Potentials of Conventional and Large Language Models on Medical Data*. *IEEE Access*, 12, 103473–103487, 2024. DOI: 10.1109/ACCESS.2024.3428918.
- [J1] **T. Sultan**, M.A.T. Rony, M.S. Islam, S. Aldosary, and W. El-Shafai. *MemesViTa: A Novel Multimodal Fusion Technique for Troll Memes Identification*. *IEEE Access*, 12, 177811–177828, 2024. DOI: 10.1109/ACCESS.2024.3505614.

Conference Papers (Published / Accepted)

- [C10] **T. Sultan**, P. Sangani, K. Cool, H.A. Akbarpour, G. Liu, and M. Babaiasl. *NeuroCommitSSM: Commit-as-Decision Readiness for Feasibility-Grounded Shared Autonomy in Assistive Robot Skills*. Accepted in the *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2026)*, 2026.
- [C9] G. Liu, N. Hawkins, B. Madden, **T. Sultan**, F. Esposito, and M. Babaiasl. *A Multimodal Data Collection Framework for Dialogue-Driven Assistive Robotics to Clarify Ambiguities: A Wizard-of-Oz Pilot Study*. Accepted in the *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2026)*, 2026. *arXiv preprint arXiv:2601.16870*.
- [C8] **T. Sultan**, K. Cool, G. Liu, G. Tamilselvan, and M. Babaiasl. *TriSaFe-Trans: A Safety-Aware Multimodal Intent Recognition Pipeline for Assistive Robotics*. Accepted in the *11th IEEE RAS/EMBS International Conference on Biomedical Robotics and Biomechatronics (BioRob 2026)*, 2026.
- [C7] G. Liu, N. Hawkins, B. Madden, **T. Sultan**, and M. Babaiasl. *A Dialogue-Based Human-Robot Interaction Protocol for Wheelchair and Robotic Arm Integrated Control*. Accepted in the *11th IEEE RAS/EMBS International Conference on Biomedical Robotics and Biomechatronics (BioRob 2026)*, 2026. *arXiv preprint arXiv:2602.06243*.
- [C6] G. Liu, **T. Sultan**, V. Di Giorgio, N. Hawkins, F. Esposito, and M. Babaiasl. *WheelArm-Sim: A Manipulation and Navigation Combined Multimodal Synthetic Data Generation Simulator for Unified Control in Assistive Robotics*. Accepted in the *International Symposium on Medical Robotics (ISMR 2026)*, 2026. *arXiv preprint arXiv:2601.21129*.
- [C5] **T. Sultan**, H.A. Akbarpour, W. El-Shafai, M. Saib, M.K.B. Bhuiyan, M.S. Islam, *et al.* *Meme-FusionNet: A Cross-Linguistic Multimodal Model for Identifying Troll Memes*. In *2025 International Conference on Control, Automation and Diagnosis (ICCAD)*, pp. 1–6, 2025.
- [C4] **T. Sultan**, M.M.H. Sarker, M.K.B. Bhuiyan, M. Saib, M.S. Islam, M.R. Hossain, *et al.* *ViTDe-BERTaNews: A Comparative Study of Single-modal, Multimodal, and LLM Techniques for Detecting Fake News*. In *2025 International Conference on Control, Automation and Diagnosis (ICCAD)*, pp. 1–6, 2025.
- [C3] J.A. Joby, P. Sikorski, **T. Sultan**, H. Akbarpour, F. Esposito, and M. Babaiasl. *EMG-TransNN-MHA: A Transformer-Based Model for Enhanced Motor Intent Recognition in Assistive Robotics*. In *2024 IEEE International Conference on Big Data (BigData)*, pp. 8691–8693, 2024.
- [C2] **T. Sultan**, G.F. Lukman, and J.W. Ahn. *Preliminary Design and Considerations of Switched Reluctance Machine for FESS Application*. In *Proceedings of the KIEE Conference*, pp. 1254–1255, 2022.
- [C1] **T. Sultan**, G.F. Lukman, K.I. Jeong, and J.W. Ahn. *Control of Switched Reluctance Motor/Generator for FESS*. In *Proceedings of the KIEE Conference*, pp. 137–139, 2022.

TRAVEL GRANTS & AWARDS

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| IEEE Robotics and Automation Society Travel Member Support Award | IROS 2026 |
| <i>Competitive travel and accommodation support</i> | 2026 |
| – Awarded reimbursement of up to \$1,500 for eligible airfare, accommodation, and travel expenses associated with attending the IEEE/RSJ International Conference on Intelligent Robots and Systems. | |
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Dean’s Travel Award — Saint Louis University |
BioRob 2026 |
| <i>School of Science and Engineering</i> | 2026 |
| – Awarded up to \$350 to support travel for presenting research at the IEEE RAS/EMBS International Conference on Biomedical Robotics and Biomechatronics. | |

Dean's Travel Award — Saint Louis University

ISMR 2026

School of Science and Engineering

2026

- Awarded up to \$350 to support travel for presenting research at the International Symposium on Medical Robotics.

IROS Travel Award — Saint Louis University

IROS 2026

Travel support for conference presentation and attendance

2026

- Awarded institutional travel support to attend and present research at the IEEE/RSJ International Conference on Intelligent Robots and Systems.